

Vibration of Structures (ME60428)

Spring 2025-26

Assignment 1 – 20 marks

1. A homogeneous steel bar is fixed at the left end, and sprung at the right end with a spring constant k , as shown below:
 - (a) Using the Hamilton's principle, derive the equation of motion, and the boundary conditions of the system.
 - (b) Using modal analysis, determine the first four eigenfrequencies and plot the corresponding mode-shape functions.
 - (c) Discuss what happens when $k \rightarrow 0$ and $k \rightarrow \infty$.

The derivations and final answers should be hand-written. You may use any software for solving the characteristic equation and plotting the mode-shapes. The code and results must be attached with the written assignment. Take $l = 2$ m, $E = 200$ GPa, $\rho = 7850$ kg/m³, $k = 5000$ Nm. The total mass of the bar is 5 kg.

Plagiarism is highly discouraged!!

